Metric-Driven Segmentation and a Frozen Co-Visibility Prior for Thoracic Lymphadenectomy

TIGER SQ-AI 2026 submission report

Omar Choudhry
Artificial Intelligence in Medicine and Surgery Group, School of Computer Science,
University of Leeds, Leeds LS2 9JT, United Kingdom
O.Choudhry@leeds.ac.uk ORCID 0000-0003-4434-3550
Team AIMS Lab · Tasks 1, 2 and 3

1 Introduction

TIGER SQ-AI covers the thoracic field during lymphadenectomy for minimally invasive esophagectomy. Tasks 1 and 2 are semantic segmentation of that field at two label granularities, 16 merged classes and 31 fine classes. Task 3 asks which of 14 lymph node stations are visible in a frame. We treat the segmentation pair and the visibility task separately, because the two are scored on different quantities and the scoring is what determines the design.

Both segmentation tasks rank by weighted Dice and weighted normalised Hausdorff distance under clinical tiers $w \in \{3, 2, 1\}$. The triple-weighted tier holds the recurrent laryngeal nerves, subclavian arteries, pulmonary artery, lymph nodes and inferior pulmonary ligaments, which are the thin, low-contrast structures a surgeon must not injure. Dice also assigns a class the score 0.0 as soon as one false component appears on a frame where that class is genuinely absent. Those two properties constrained the method more than any architectural choice available to us.

2 Tasks 1 and 2: segmentation

2.1 Architecture and losses

The backbone is nnU-Net v2 [1] in its 2D configuration, since every input is a single video frame and no temporal context is provided at inference. The two tasks use different losses.

Task 1, over 16 merged classes, uses TopK cross-entropy restricted to the hardest 10% of pixels, combined with a Tversky term [2, 4]. Task 2, over 31 fine classes, uses cross-entropy combined with Focal-Tversky [3] at α=0.3, β=0.7, $\gamma \approx 1.33$; three seeds are trained and their probability maps averaged before a single argmax.

The split is a measured result rather than a preference. The same recall-weighted Focal-Tversky loss improved one task and damaged the other: on Task 2 it raised w=3 Dice from 0.464 to 0.498, while on Task 1 it lowered the same quantity from 0.349 to 0.273. Task 1's weighted groups are unions of comparatively large regions, so a loss that pushes recall over-predicts them and Dice penalises the excess. Hard-pixel mining suits that setting instead, and it raised Task 1's hardest class, the nerves, from 0.243 to 0.500.

2.2 Two constraints the data imposes

Mirroring is disabled in training and at test time. Thoracic anatomy is laterally asymmetric and the left and right structures are separate classes, so a horizontal flip maps the right recurrent laryngeal nerve onto the left. The augmentation that nnU-Net applies by default would therefore invert labels systematically on exactly the w=3 tier. This is a domain constraint, not a compute-budget one.

False positives are suppressed with a per-class connected-component area filter that requires no training. On frames where a class is truly absent, our false components are small, at most 28 px, while genuine masks of the same class exceed 116 px, so a single area threshold separates them with no measured loss of true detections. Given that Dice zeroes an absent class on one spurious component, this filter is worth more than its simplicity suggests.

2.3 Two refuted upgrades, and what they implied

Two upgrades aimed directly at the thin-structure bottleneck were tested and rejected: a high-resolution cascade with topology-aware losses, which cost 0.047 w=3 Dice, and a promptable refiner built on SAM 2.1 [5], which returned 0.003. When two independent methods that target the same bottleneck both fail, the most economical reading is that the limit is imposed by the training data rather than recoverable by the method. The models available at that point had seen 140 frames from a single centre, which contain very few instances of a recurrent laryngeal nerve.

The organisers' release of 31 August, comprising 40 cases and 524 frames across six centres, tested that reading directly. We retrained with the recipe held fixed, so that any change in score isolates the effect of data from the

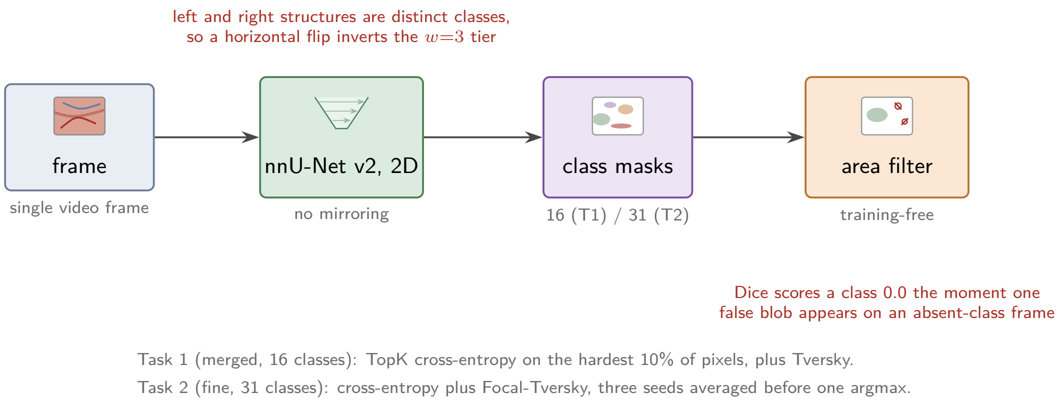

Figure 1: Segmentation pipeline for Tasks 1 and 2. A single frame is segmented by a 2D nnU-Net with mirroring disabled, and the predicted class masks pass through a training-free per-class area filter before scoring. The two tasks share this arrangement and differ only in label granularity and loss.

effect of method.

3 Task 3: lymph node station visibility

3.1 Two branches

The first branch is conditioned on our own segmentation. From the Task-2 output we read presence and log-area for 18 boundary structures, giving 36 features, and map them through a small MLP to 14 station logits. The annotation protocol defines a station as in frame when the structures surrounding it are visible, so the segmentation output is the natural evidence for it.

The second branch is a 14×14 co-visibility matrix holding Pr(station s visible | station a), estimated from training-label counts with additive shrinkage at α=2.0, fixed in advance, toward the pooled marginal. It has no learned parameters and needs no GPU. The two branches are blended at a fixed $\lambda = 0.5$.

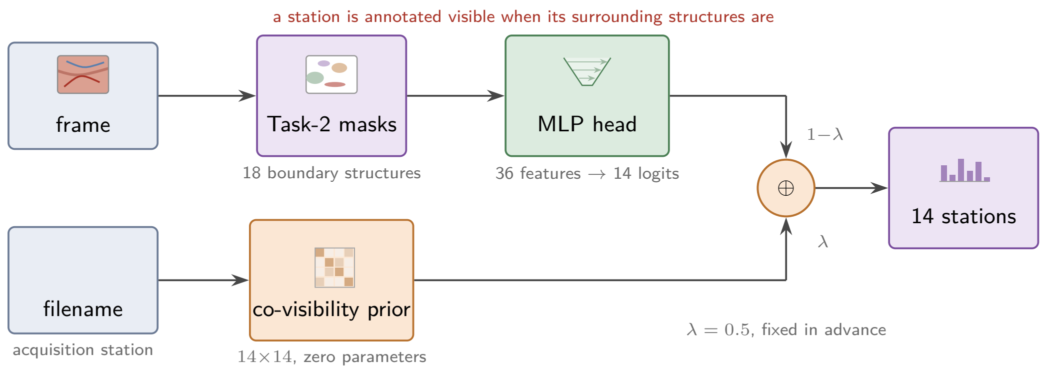

Figure 2: Station visibility. The upper branch derives features from the Task-2 masks; the lower branch looks up a co-visibility row for the acquisition station named in the input filename. The weight $\lambda = 0.5$ was fixed before any evaluation and was not tuned afterwards.

3.2 Disclosure

The prior is keyed on the acquisition station named in the input filename. The challenge data contract specifies that filename format and guarantees it is delivered unchanged, so this is documented input metadata rather than inspection of test labels. It is nevertheless not pixel evidence, and reviewers should be able to see that clearly. On our held-out protocol the image-only head reaches macro-F1 0.6276 and the submitted blend reaches 0.7128, a gain of 0.085. If the organisers rule the filename out of scope, the image-only model stands at 0.6276.

3.3 Why the submitted configuration is not our highest-scoring one

The prior alone scores 0.7307, above the blend we submit. We submit the blend because $\lambda = 0.5$ was fixed before any measurement was taken, and moving it after seeing that number would be selection on the evaluation set.

The multi-centre release supports that decision on its own terms. The matrix was fitted on 10 cases from a single centre. Evaluated leave-one-centre-out on the new release, macro-F1 falls to 0.5998, and the worst pairwise drift between per-centre matrices is 0.3737, well past the 0.15 threshold at which our pre-registered rule declares the prior centre-specific and prefers the blend. The higher-scoring configuration was the more brittle one.

A second trade-off is worth reporting because it does not resolve. On 298 frames from centres the matrix has never seen, the single-centre matrix ranks well but is poorly calibrated at the 0.5 threshold: AUROC 0.903, macro-F1 0.507. Refitting on all 40 cases repairs the threshold and damages the ranking: macro-F1 0.597, AUROC 0.801. The task is scored on both quantities, so neither configuration dominates, and the frozen matrix was kept under the pre-registered rule.

4 Results

Table 1 reports the retrained configuration. The evaluation set is 75 held-out frames drawn from centres 2, 3, 4, 6 and 7, none of which the incumbent models had seen in training, scored with the organisers' code.

task	quantity	incumbent	retrained
Task 2 (fine, 31 classes)	w=3 Dice	0.3733	0.7372
from multi-centre data			+0.3253
from area filtering			+0.0386
Task 1 (merged, 16 classes)	w=3 Dice	retrain −0.0140, incumbent retained	
Task 3 (14 stations)	macro-F1	+0.1152 from the improved masks	

Table 1: Effect of retraining on the multi-centre release, measured on 75 held-out frames from five centres absent from the incumbent's training data. The Task-2 gain is decomposed into the part attributable to data and the part attributable to the area filter.

The submitted configuration is therefore the retrained three-seed Task-2 ensemble with the area filter, the unchanged Task-1 model, and the unchanged Task-3 blend.

Earlier internal figures of 0.552 for Task 1 and 0.702 to 0.709 for Task 2 were measured on the single-centre split available before the August release. They differ from Table 1 in evaluation data, not only in model, and the two should not be read alongside each other.

5 Data disclosure

Provided challenge data only. The pretrained components are nnU-Net's standard initialisation. A diffusion-synthesised augmentation for one Task-2 class was trained from the provided training data alone and evaluated, but it is not part of the submitted configuration. No private data, no additional non-public annotations, and no test-derived quantity enters any hyper-parameter. Training and inference code is public under MIT at `https://github.com/omariosc/tiger-aims`.

6 Limitations

No official score exists at the time of writing, so every figure here is our own measurement using the organisers' scoring code, on splits we constructed.

Task 3 depends on filename metadata, disclosed above, and its matrix is single-centre in origin. Refitting repairs calibration at the cost of ranking, and choosing between those requires knowing how the two published metrics combine. The organisers publish macro-F1 and AUROC for Task 3 but not that rule, so taking the mean of the two is our own aggregation, chosen in advance for neutrality. This is an open risk rather than a formality: if the ranking is macro-F1 alone, the refit we rejected would have scored 0.090 higher.

References

[1] F. Isensee, P. F. Jaeger, S. A. A. Kohl, J. Petersen, K. H. Maier-Hein. nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation. *Nature Methods*, 18(2):203–211, 2021.

[2] S. S. M. Salehi, D. Erdogmus, A. Gholipour. Tversky loss function for image segmentation using 3D fully convolutional deep networks. *MLMI*, 2017.

[3] N. Abraham, N. M. Khan. A novel focal Tversky loss function with improved attention U-Net for lesion segmentation. *ISBI*, 2019.

[4] Z. Wu, C. Shen, A. van den Hengel. Bridging category-level and instance-level semantic image segmentation. arXiv:1605.06885, 2016.

[5] N. Ravi, V. Gabeur, Y.-T. Hu, et al. SAM 2: Segment Anything in Images and Videos. arXiv:2408.00714, 2024.